

eQTL Quality Control, Candidate Prioritization & Trans-Band Analysis

Adipose & Liver Tissues from HSNiH-Palmer Rat Cohort (GeneNetwork)

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Outline

- 1 Data & Pipeline Overview
- 2 Key Finding 1: Keep the Trans Hits
- 3 Key Finding 2: Trans-Bands
- 4 Key Finding 3: Cross-Tissue Cis Conservation
- 5 Divergence from Hong-Le et al. 2023
- 6 Next Steps & Proposals

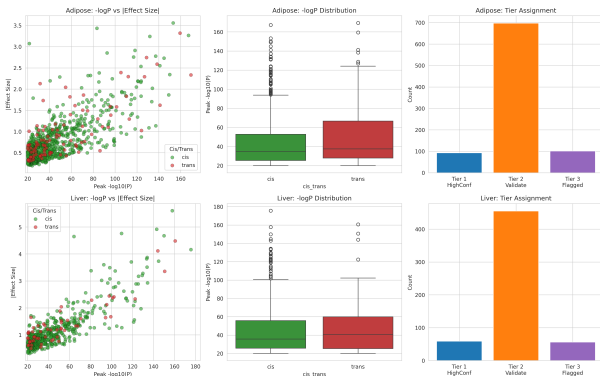
What We Started With

Input: Pre-filtered eQTL tables from GeneNetwork 2 ($-\log_{10} P \geq 20$)

- **Adipose:** 891 genes
- **Liver:** 570 genes
- Per-gene best peak: location, $-\log P$, effect size

Pipeline:

- 1 Cis vs Trans classification (4 Mb window)
- 2 Flag trans hits (MEGA_EFFECT, PSEUDOGENE, LOC, CROSS_TISSUE)
- 3 Three-tier scoring (Tier 1 / 2 / 3)
- 4 Cross-tissue integration + master scoring



Why This Matters: 33 Cross-Tissue Trans Hotspots

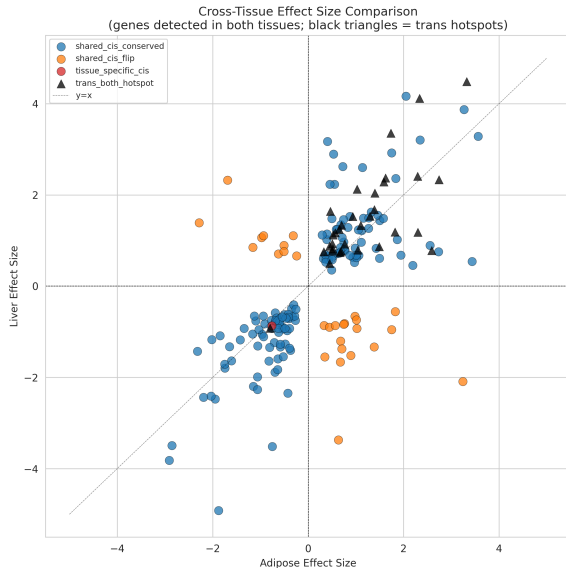
If a trans signal replicates in both adipose and liver, it is unlikely to be random noise.

What we changed:

- **Before:** Trans-eQTLs flagged as “artifacts” and excluded
- **After:** All 157 trans hits retained with 4 flags

Possible biological explanations:

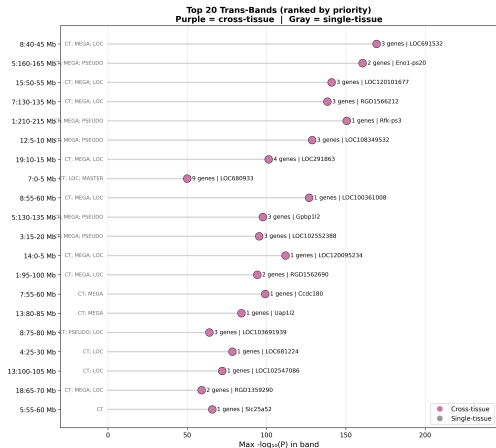
- 1 **True master regulator** — one variant drives many distal genes
- 2 **Copy-number variant (CNV)** or segmental duplication
- 3 **Mapping ambiguity** from a linear reference lacking founder-specific sequence



Black triangles = trans_both_hotspot genes

Trans-Bands: Genomic Hotspots of Distal Regulation

Definition: A 5 Mb window where multiple independent genes carry a trans-eQTL peak.

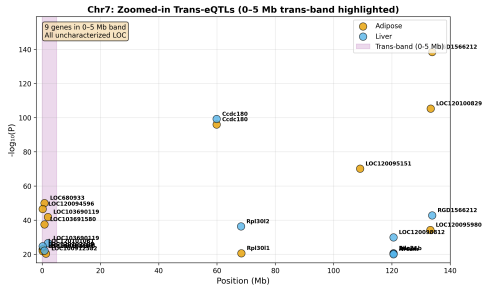
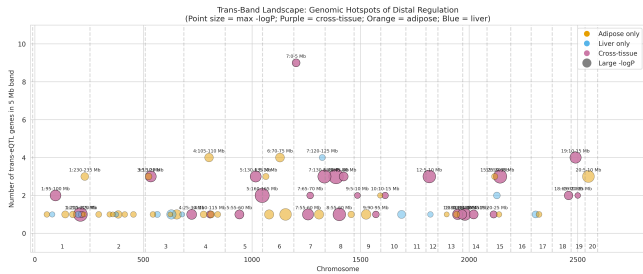


Top bands (all cross-tissue):

- **Chr8:40–45 Mb** — 3 genes, max $-\log P = 169.4$ (LOC691532)
- **Chr5:160–165 Mb** — 2 genes, max $-\log P = 160.6$ (Eno1-ps20)
- **Chr4:105–115 Mb** — 6 genes, max $-\log P = 59.4$
- **Chr7:0–5 Mb** — 9 genes (telomeric cluster)

Proposal: These are candidates for pangenomic/CNV inspection, not artifacts.

Chr7:0–5 Mb — The Strongest Trans-Band

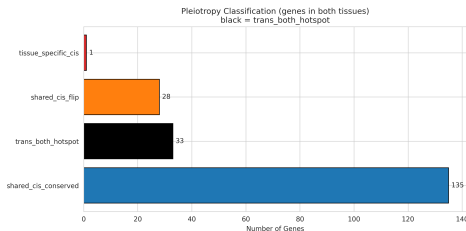


Why Chr7 is interesting:

- 9 genes map to the very start of Chr7 (within 5 Mb of the telomere)
- All are uncharacterized LOC genes
- Could be a genuine regulatory cluster, an assembly gap, or a segmental duplication
- **Action item:** Check RGD for known CNVs; inspect raw read depth

Most Strong Cis-eQTLs Are Conserved Across Tissues

197 genes detected in both tissues:



Class	Count
Shared conserved cis	135
Shared cis, flipped	28
Trans in both (hotspot)	33
Tissue-specific cis	1

Take-home: 69% of shared genes have the **same cis peak, same direction** in both tissues. The genetic architecture of expression is largely shared between adipose and liver.

What We Did Differently From the Paper

Aspect	Hong-Le et al.	Our Analysis
Genome coverage	18K / 17K transcripts (FDR $\leq$ 0.001)	Pre-filtered tail ($-\log P \geq 20$)
Cis window	1 Mb	4 Mb (captures distal enhancers in TADs)
Trans-eQTLs	Essentially ignored	Retained, flagged, clustered
Annotation focus	Well-known mediators (Grk5, Krtcap3)	40% uncharacterized LOC/pseudogenes

This is not a bug; it's a different lens on the same data. The paper asked “which cis-eQTLs mediate metabolic traits?” We ask “what systematic genomic features are hiding in the trans signal?”

Two-Track Path Forward

Track 1: Cis Candidates

- Manual curation (verify symbols, check peaks in RGD/Ensembl)
- Colocalization with metabolic GWAS (COLOC)
- Fine-mapping on top 5–10 loci

Track 2: Trans-Bands

- Raw read-depth analysis at top 3 bands
- Founder haplotype tracing (8 HS founders)
- **Pangenomic re-mapping:**
Minigraph-Cactus with founder assemblies

Suggestive hypothesis: Population-segregating structural variants, absent from the linear reference, act as master regulators of multiple expression traits.

- ① **All trans-eQTLs are retained** with granular flags — no more silent exclusion
- ② **7 trans-bands** identified, 3 replicating across tissues — candidates for CNV/SV inspection
- ③ **135/197** shared genes have conserved cis regulation — expression architecture is largely tissue-shared
- ④ **Intentional divergence** from Hong-Le et al.: we focus on the extreme tail and the trans signal
- ⑤ **Proposal:** Pangenomic re-mapping of top trans-bands to test the “hidden structural variant” hypothesis

Thank you — Questions?